MediFlow

Reducing Wait - Enhancing Care

MediFlow

LEADER	Prabhanshu	9999720708
Member 1	Dakshesh	
Member 2	Shivansh	
Member 3	ChandraMouli	

INTRODUCTION

Hospitals struggle with long waiting times and inefficient queue management, leading to dissatisfaction and health risks. The AI-powered hospital queue management system optimizes patient flow, predicts waiting times, and prioritizes care dynamically

System Overview & Features

AI assesses patient conditions using cameras and wearable devices, predicts waiting and arrival times, and integrates with hospital queues. It updates queues in real time, dynamically prioritizing patients based on urgency for better efficiency and experience.

Long waiting times and inefficient queue handling lead to patient frustration and delays in urgent care

Endless Hospital Queues

Long Waits, Rising Frustration!

Inefficient Hospital Scheduling

ISSUES

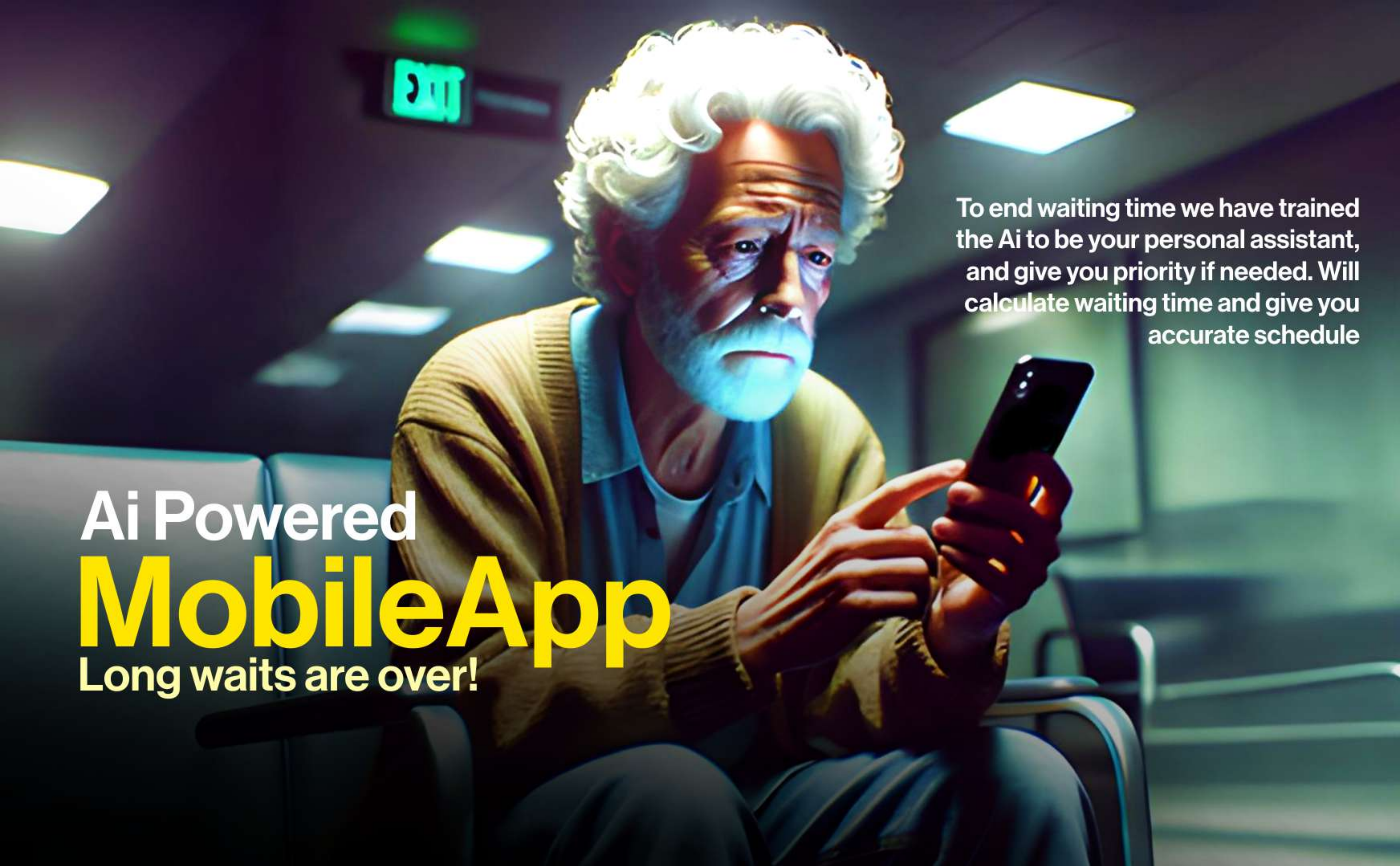
• Long waiting times, last-minute cancellations, and poor doctor-patient matching.

IMPACT

• Overburdened staff, patient dissatisfaction, and resource mismanagement.

SOLUTION

• AI-based smart scheduling to optimize appointments and hospital



To end waiting time we have trained the Ai to be your personal assistant, and give you priority if needed. Will calculate waiting time and give you accurate schedule

Ai Powered
MobileApp
Long waits are over!

Poor appointment management results in long waits, last-minute cancellations, and staff inefficiencies, affecting both patients and doctors.

Chaotic Appointments

No-Shows, Delays & Mismanagement!

AI-Driven Scheduling

1) AI-Based Triage: Analyzes patient symptoms & prioritizes cases.
2) Automated Scheduling: Matches patients with doctors based on urgency & availability.
3) Dynamic Adjustments: AI reschedules appointments instantly for no-shows.
4) Resource Optimization: AI predicts bed occupancy, staff needs & equipment demand.

Workflow

1. AI evaluates patient condition & history.
2. AI assigns appointment priority.
3. AI dynamically adjusts for delays.
4. AI optimizes hospital resources.

Packed Waiting

ROOMS

Overflowing Patients, Limited Resources!

Packed waiting rooms, lack of real-time resource allocation, and limited staff lead to inefficiencies in patient care

How Will We Train AI for Scheduling?

Data Sources for AI Training

Historical hospital data

• Past appointments, doctor availability, patient records.

Real-time patient flow data

• Live check-ins, cancellations, waiting times.

Emergency vs. routine cases

• AI learns to categorize and prioritize cases.

Staff workload & hospital capacity


Ambulance
Delays
Traffic & Overcrowded ERs
Cost Lives!
Traffic congestion, outdated routing, and overcrowded ERs slow emergency response, putting critical patients at risk

Slow Emergency Response

• Issues: Traffic congestion, hospital overcrowding, outdated routing methods.
• Impact: Delays in critical care, increased mortality rates.

Workflow

AI-driven ambulance system for real-time routing & hospital coordination.

How AI Improves Ambulance Efficiency

· Traffic Signal AI: Ambulance gets green-light priority
· Smart Route Selection: AI finds the fastest path using real-time traffic data
· Emergency Classification: AI determines if ER is needed or home care is better
· Live Hospital Dashboard: Ambulances get real-time updates on hospital capacity

AI Workflow

1. AI assesses emergency severity.
2. AI chooses the best hospital based on bed availability.
3. AI clears traffic paths by linking with smart signals.4. AI updates the ER team for immediate patient care.


Ai powered
Ambulance
Personal Doctor in an ambulance
This personal ai will help the ambulance staff to help the paitent with a larger knowledge base

Train AI for Ambulance Routing

Data Sources for AI Training:

- GPS & Traffic Data: Real-time congestion patterns.
- Emergency Response Data: Historical response times & outcomes.
- Hospital Capacity Data: Bed occupancy, staff availability.
- Ambulance Movement Data: Speed, efficiency, past routes.

Training Process:

1. Data Collection: Use real-time traffic & emergency response logs
2. Model Training: AI learns patterns in traffic flow, hospital load, & emergency urgency
3. Simulation Testing: AI runs scenarios to determine optimal ambulance routes
4. Continuous Learning: AI adapts based on live performance feedback

Traffic congestion, outdated routing, and overcrowded ERs slow emergency response, putting critical patients at risk

Undetected Infections

A Silent Threat in Hospitals!

Thermal AI Cameras

Enhancing Healthcare with AI-Powered Thermal Cameras

Why Thermal AI Cameras?

- Detects body temperature variations in real time.
- Identifies fever, infections, or critical conditions without physical contact.
- Helps in early diagnosis of illnesses like COVID-19, flu, and infections.
- Reduces hospital-acquired infections by screening patients & visitors.

Key Applications:

- Hospital Entry Screening: Detects feverish individuals before entry.
- Emergency Triage: AI identifies critical cases instantly.
- Infection Spread Control: Alerts staff about potential outbreaks.
- Remote Patient Monitoring: AI tracks temperature trends over time


Will scan the humans enterting the facility and will extract heat maps for further information about the subject
Ai Powered
SCANNING

How AI-Powered Thermal Cameras Work?

- Infrared Sensors: Capture heat signatures from individuals.
- AI Image Processing: Analyzes temperature variations across body regions.
- Smart Alerts: AI flags abnormal temperatures for medical attention.
- Integration with Hospital Systems: Sends real-time data to doctors & triage teams.

How AI is Trained for Thermal Detection?

1. Data Collection: Large datasets of body temperatures from different conditions.
2. Pattern Recognition: AI learns differences between normal & feverish patients.
3. Anomaly Detection: AI gets trained to detect unusual temperature spikes.
4. Real-Time Learning: Continuous improvements based on hospital feedback.

Ai Powered Heat Maps

As the first thing body does when infected with an virus or diesease is to change normal body temperatur

Exhausted Doctors

Healthcare on the Brink

Overworked doctors and nurses face burnout due to inefficient shift planning and unbalanced task distribution.

Staff Workload Optimization

Challenges in Staff Management

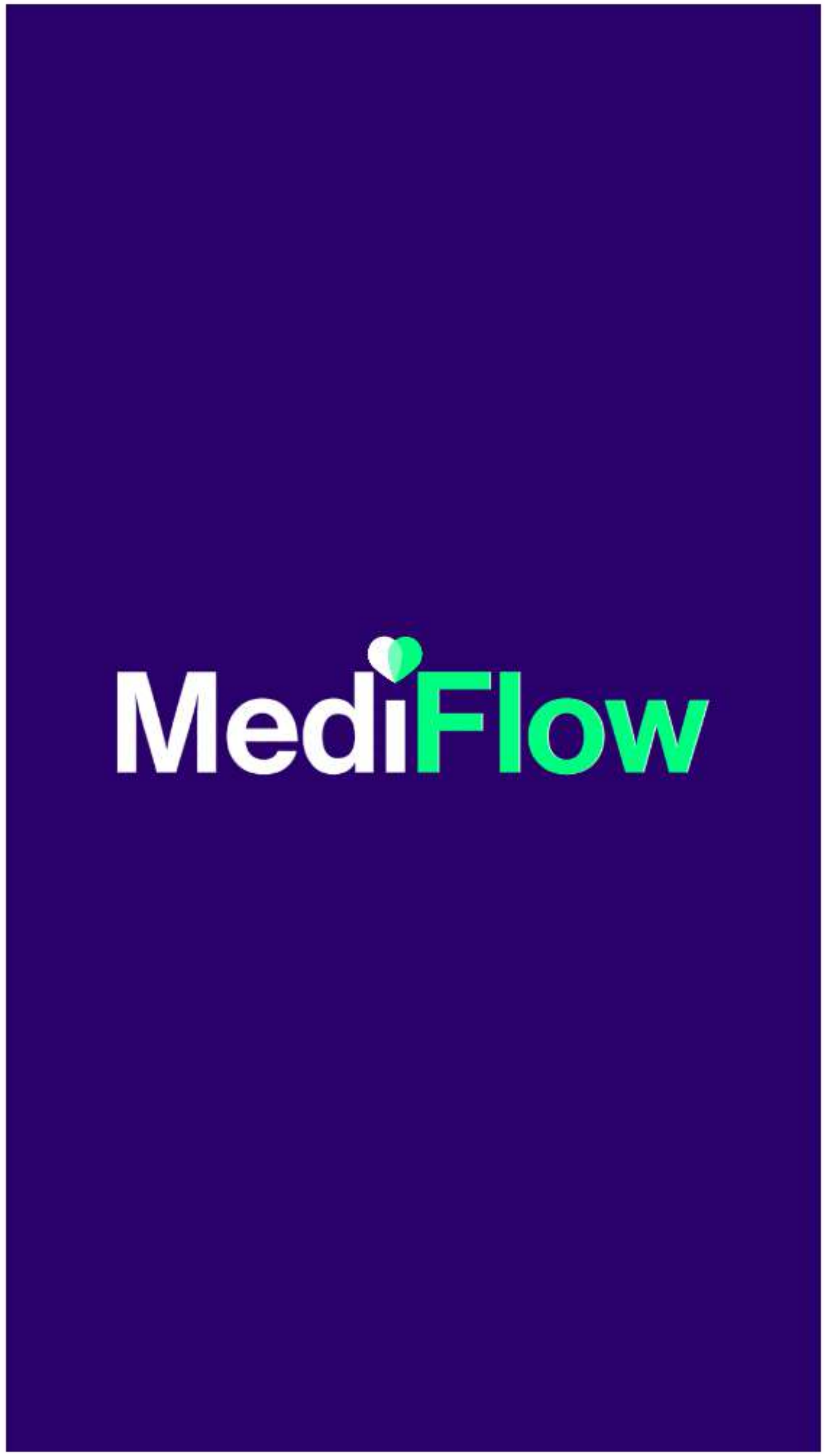
- Uneven workload   Some staff overburdened, others underutilized
- Inefficient shifts   Leads to exhaustion and reduced efficiency
- High patient influx   Real-time workload balancing is difficult
- No real-time adjustments   Static schedules don't adapt to emergencies

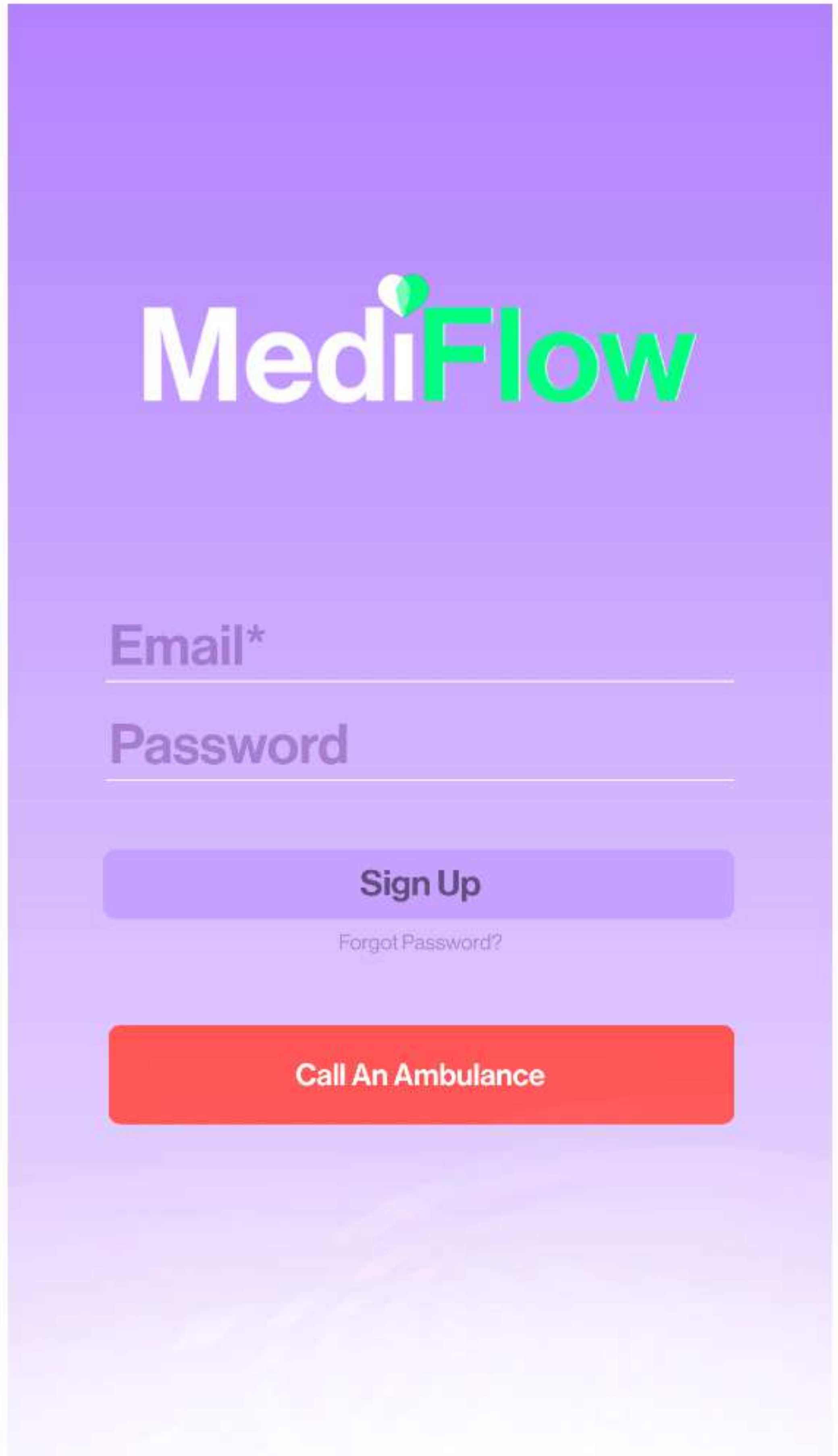
How AI Solves These Challenges

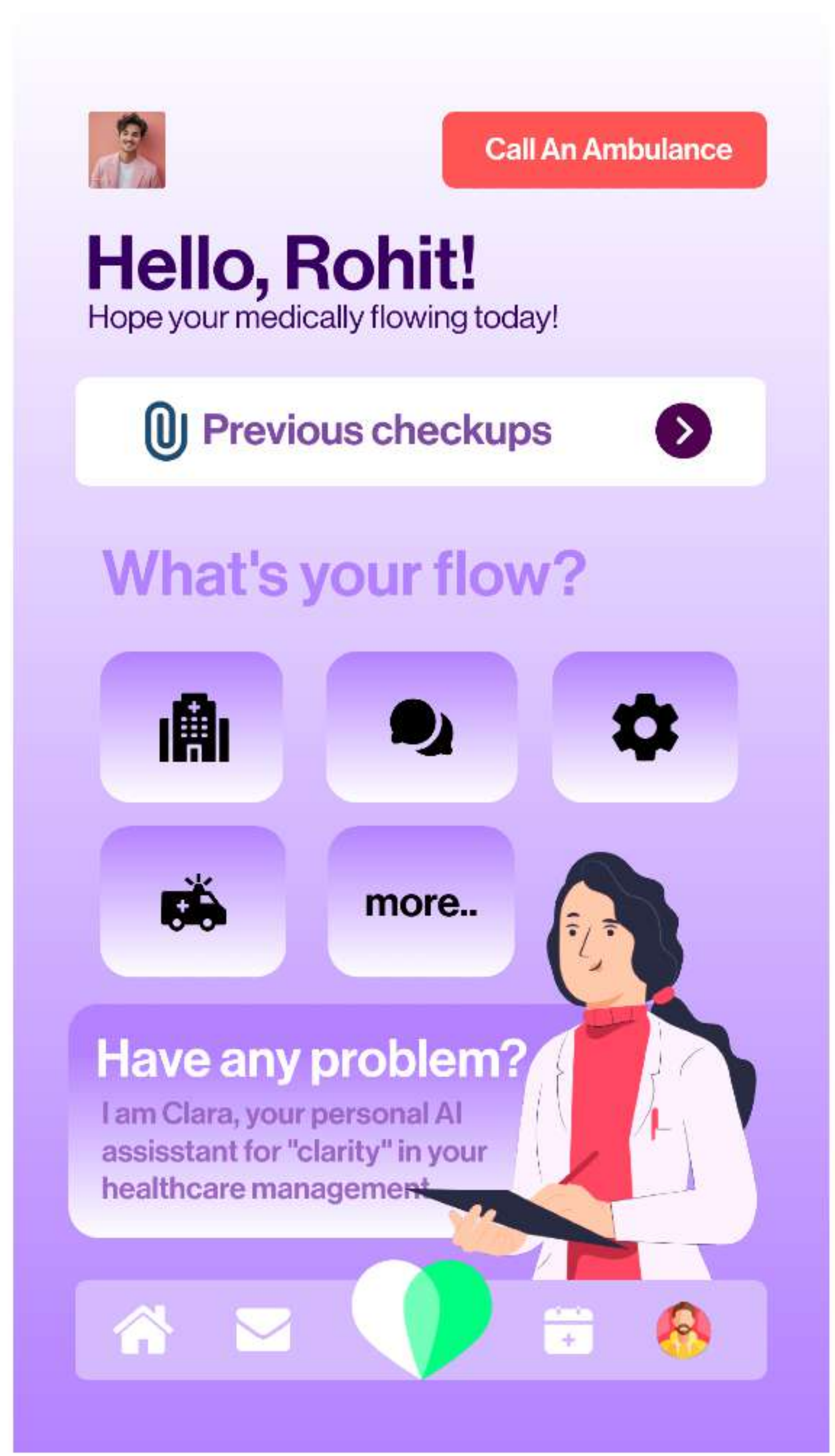
- Smart Shift Optimization   AI assigns shifts based on expertise, availability, and demand.
- Real-Time Workload Balancing   AI detects staff overload and redistributes tasks.
- Fatigue & Work Hour Monitoring   AI tracks work hours, preventing exhaustion.
- AI Dashboard for Administrators   Provides real-time insights into staff workload.

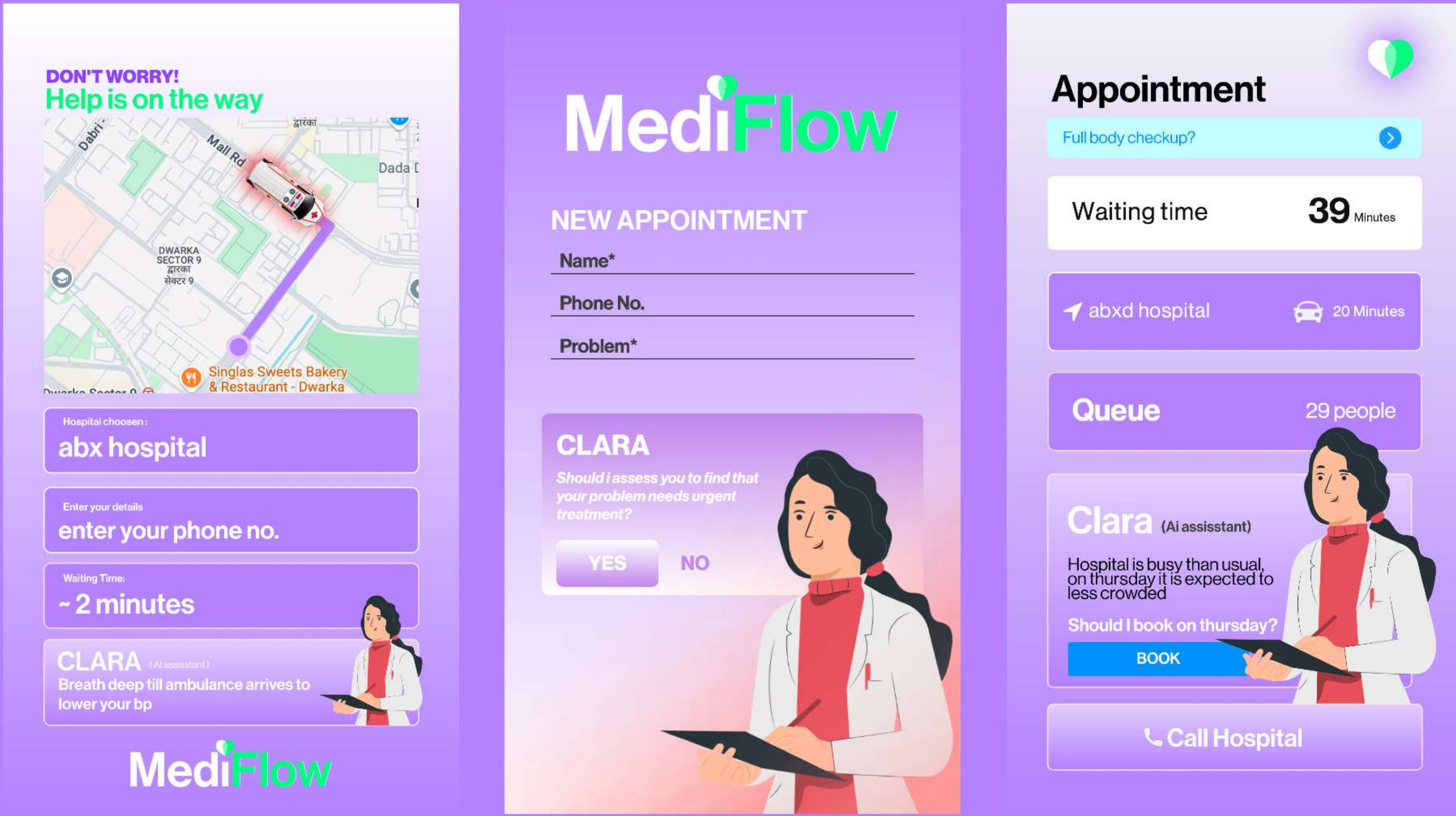
AI Workflow

- Data Collection & Learning   AI gathers real-time & historical data on staff schedules and patient flow.
- Workload Prediction & Optimization   AI predicts peak hours and suggests shift modifications.
- Dynamic Task Allocation   AI redistributes tasks to balance workloads and prevent burnout.
- Live Monitoring & Automated Adjustments   AI tracks fatigue, overtime trends, and emergency spikes.


MediFlow


MediFlow
Email*
Password
Sign Up
Forgot Password?
Call An Ambulance


Call An Ambulance
Hello, Rohit!
Hope your medically flowing today!
Previous checkups
What's your flow?
more..
Have any problem?
I am Clara, your personal AI assisstant for "clarity" in your healthcare management



DON'T WORRY!
Help is on the way
Mall Rd
DWARKA SECTOR 9
द्वारका सेक्टर 9
Singlas Sweets Bakery & Restaurant - Dwarka
Hospital choosen :
abx hospital
Enter your details
enter your phone no.
Waiting Time:
~ 2 minutes
CLARA (AI assisstant)
Breath deep till ambulance arrives to lower your bp
MediFlow
MediFlow
NEW APPOINTMENT
Name*
Phone No.
Problem*
CLARA
Should i assess you to find that your problem needs urgent treatment?
YES
NO
Appointment
Full body checkup?
Waiting time 39 Minutes
abxd hospital
20 Minutes
Queue
29 people
Clara (Ai assisstant)
Hospital is busy than usual, on thursday it is expected to less crowded
Should I book on thursday?
BOOK
Call Hospital

Summary

1. AI-Powered Queue Management
AI optimizes patient flow by assessing conditions through cameras and wearables, predicting waiting times, and dynamically prioritizing cases. Patients receive real-time notifications about their queue position.

2. AI-Driven Scheduling
AI automates appointment booking, prioritizes urgent cases, and reschedules no-shows instantly. It improves resource utilization by predicting doctor availability and hospital capacity.

3. AI for Ambulance Efficiency
AI-driven ambulances use real-time traffic data for optimal routing, coordinate with hospitals for bed availability, and receive green-light priority. AI learns from emergency response patterns to improve efficiency.

4. AI-Powered Thermal Cameras
Thermal AI cameras detect fever and infections in real time, assisting in hospital entry screening, emergency triage, and infection control. AI continuously improves by analyzing temperature

5. AI in Staff Workload Management

AI optimizes shift planning, prevents staff burnout, and dynamically redistributes tasks. It tracks fatigue levels, predicts workload spikes, and provides real-time insights for administrators.

6. AI Solutions for Overcrowding

AI predicts peak patient influx, allocates beds efficiently, and adjusts staffing dynamically. It prevents bottlenecks by balancing patient distribution across hospital departments.

7. AI Training & Continuous Learning

AI is trained using historical hospital data, real-time patient flows, and staff workload trends. It continuously learns from real-world hospital operations to enhance

Thank You